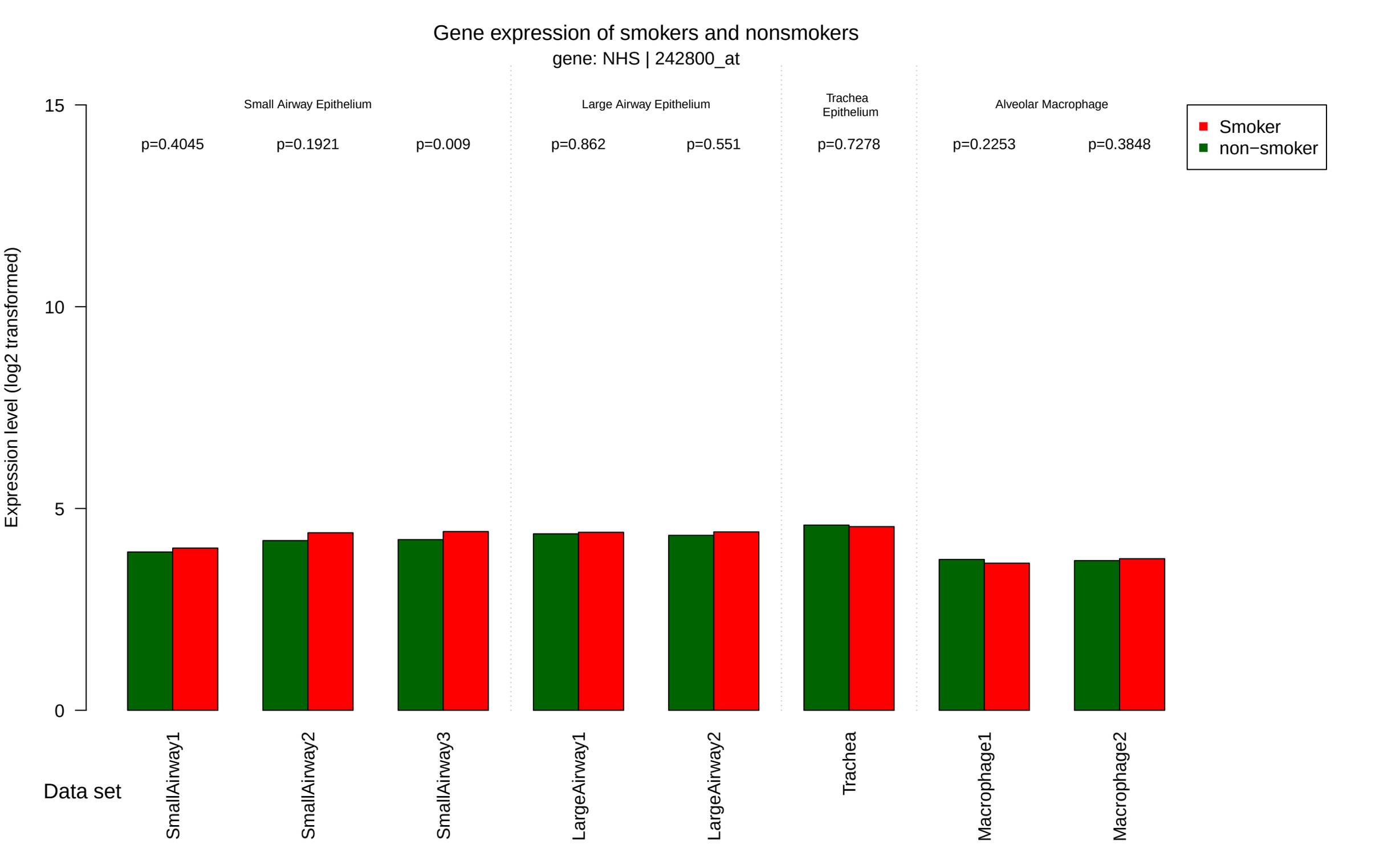
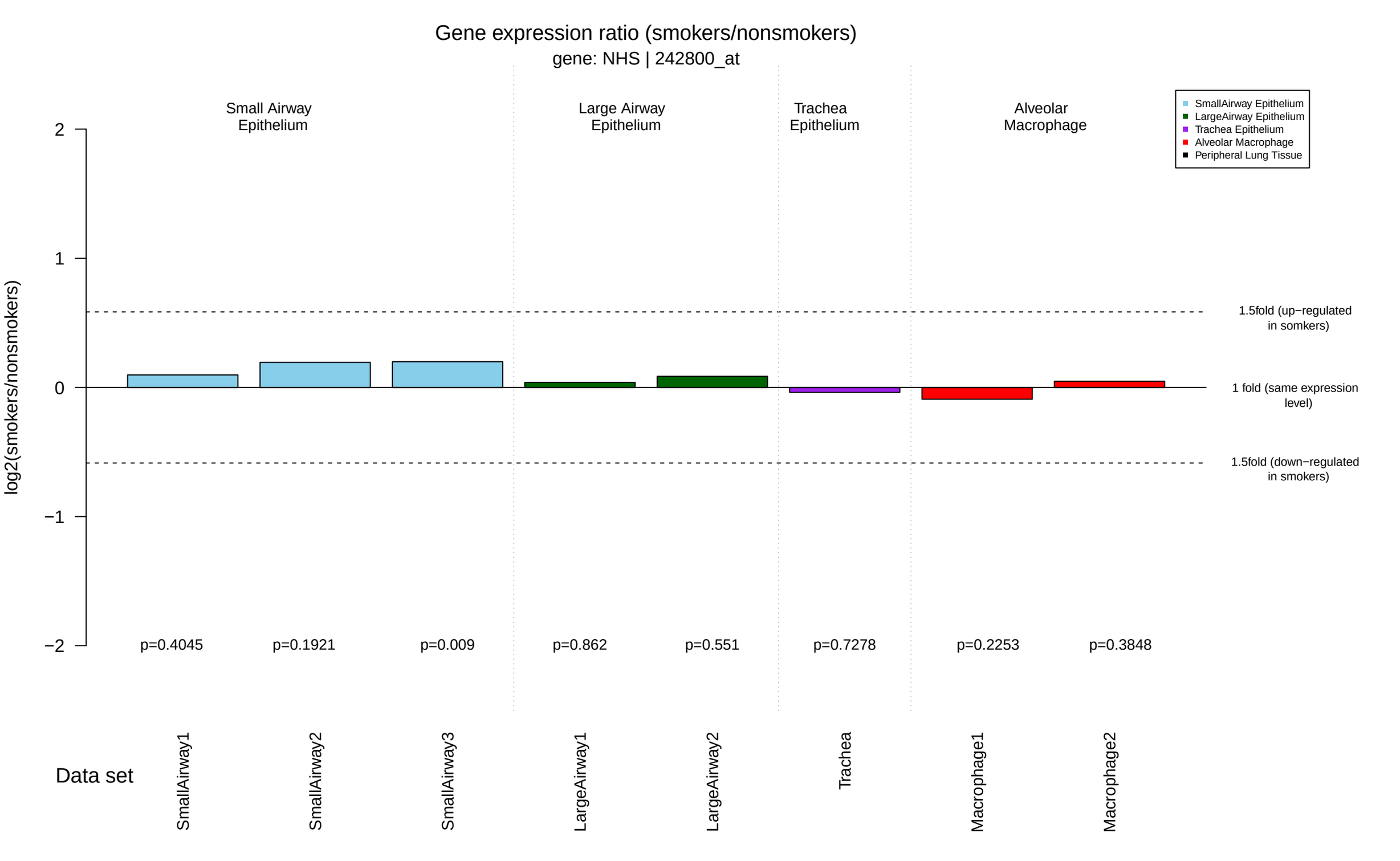
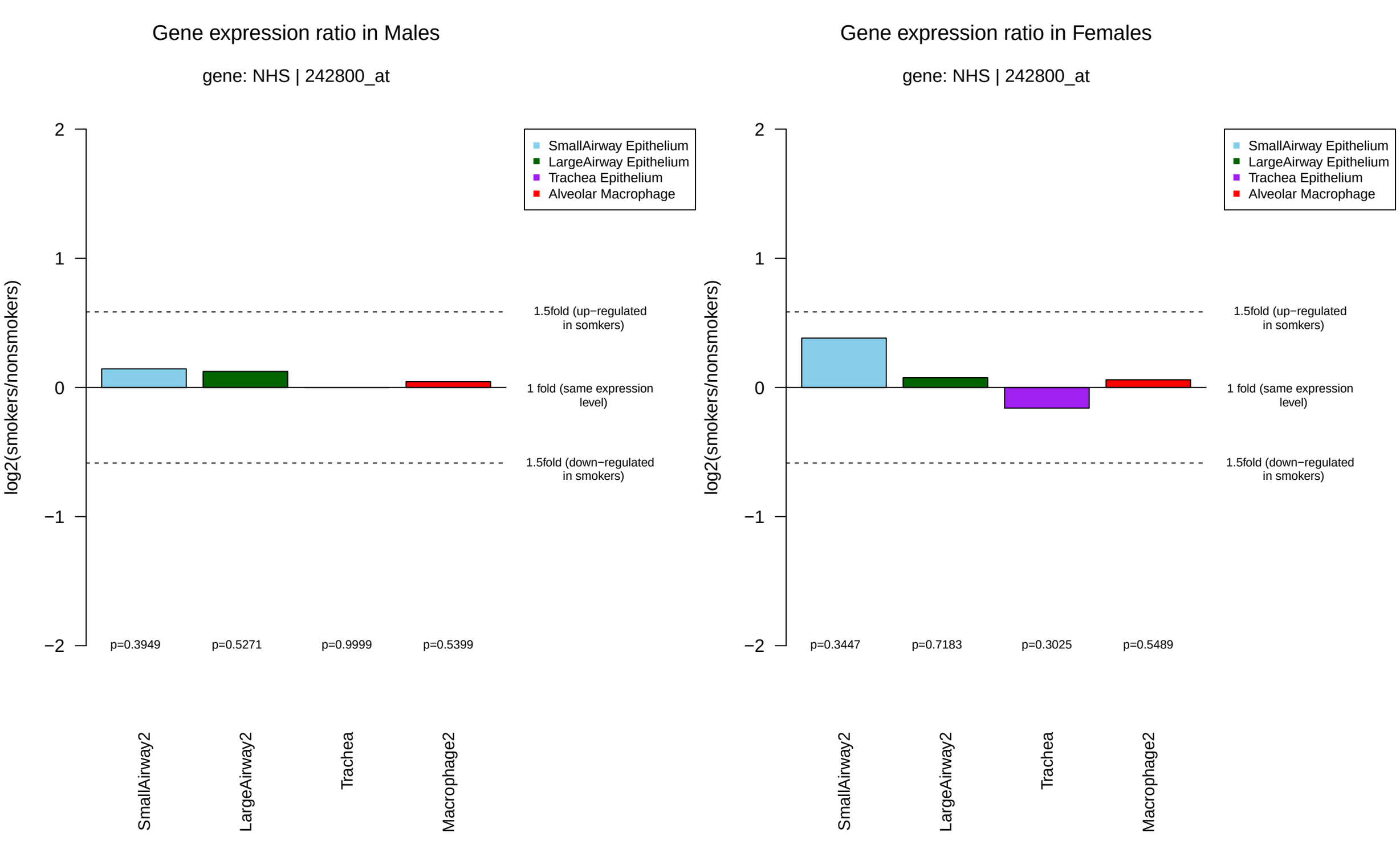
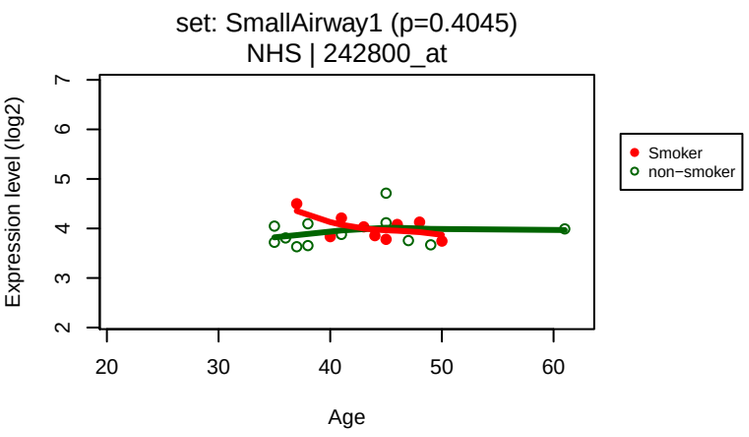
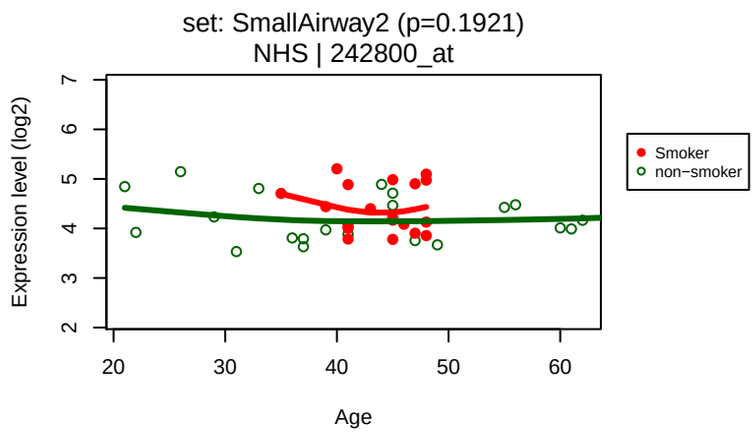
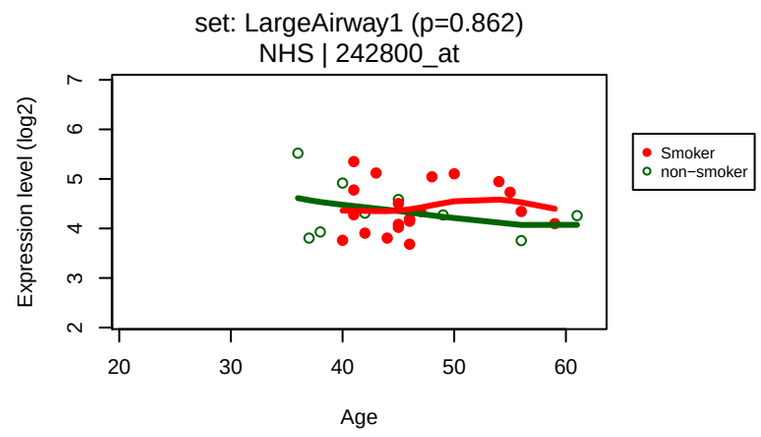

Gene expression of smokers and nonsmokers
gene: NHS | 242800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4045
p=0.1921
p=0.009
p=0.862
p=0.551
p=0.7278
p=0.2253
p=0.3848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

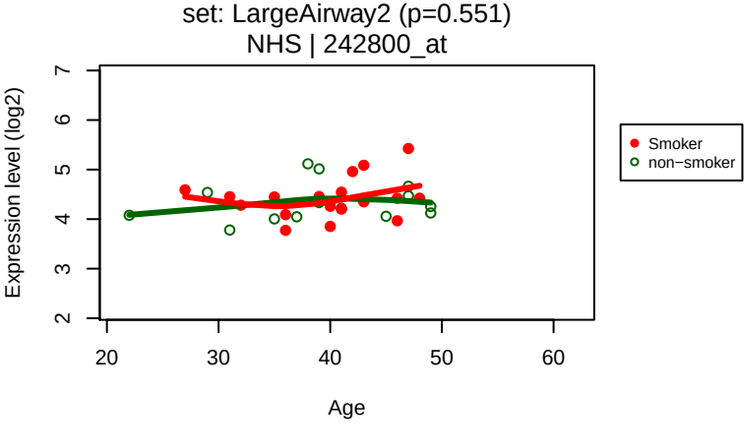

Gene expression ratio (smokers/nonsmokers)
gene: NHS | 242800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4045
p=0.1921
p=0.009
p=0.862
p=0.551
p=0.7278
p=0.2253
p=0.3848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

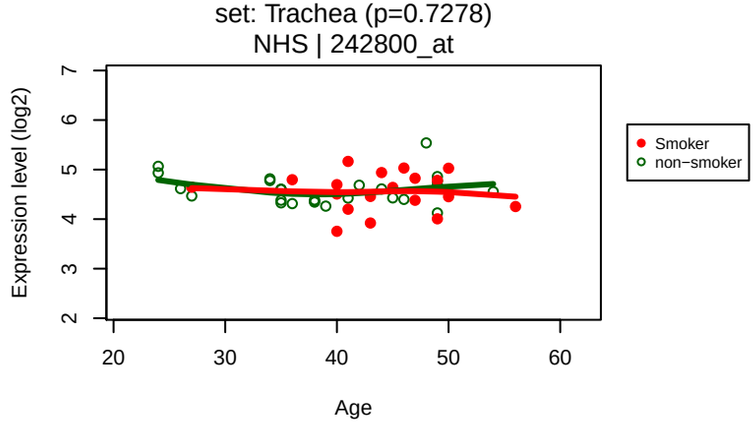

Gene expression ratio in Males
gene: NHS | 242800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3949
p=0.5271
p=0.9999
p=0.5399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NHS | 242800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3447
p=0.7183
p=0.3025
p=0.5489
SmallAirway2
LargeAirway2
Trachea
Macrophage2

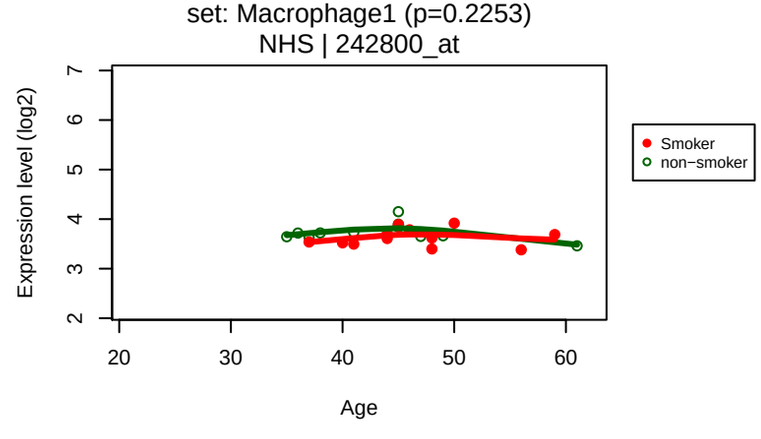

set: SmallAirway1 (p=0.4045)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker

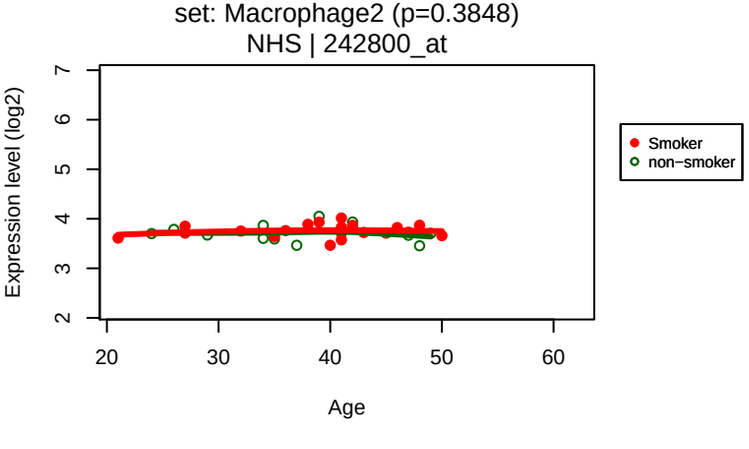

set: SmallAirway2 (p=0.1921)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker

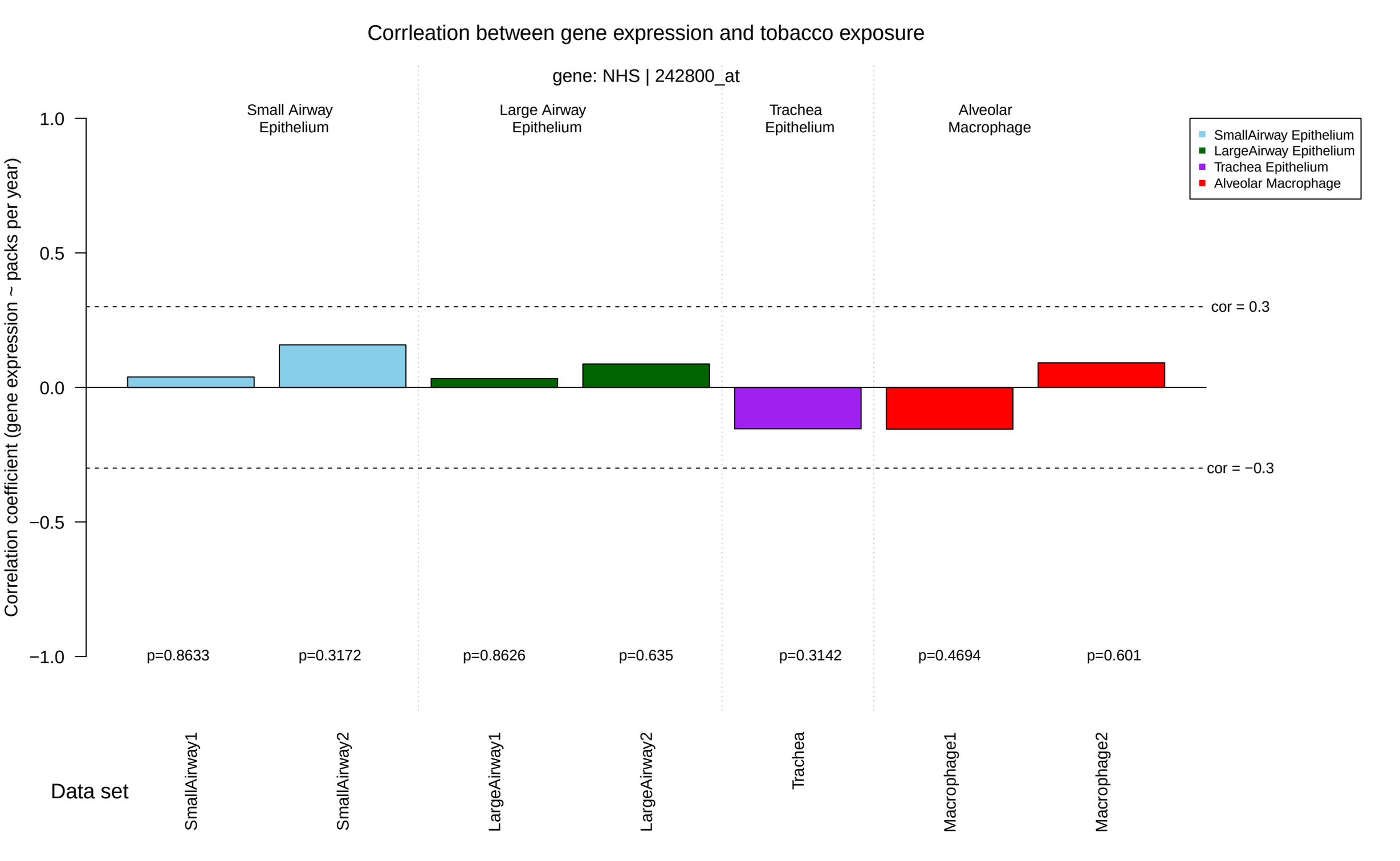

set: LargeAirway1 (p=0.862)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.551)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7278)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2253)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3848)
NHS | 242800_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NHS | 242800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8633
p=0.3172
p=0.8626
p=0.635
p=0.3142
p=0.4694
p=0.601
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2